

# Magic, Superpowers, or Empowerment?

## A Conceptual Framework for Magic Interaction Techniques

Bastian Dewitz\*  
Human-Computer Interaction  
Universität Hamburg

Sukran Karaosmanoglu†  
Human-Computer Interaction  
Universität Hamburg

Robert W. Lindeman‡  
HIT Lab NZ  
University of Canterbury

Frank Steinicke§  
Human-Computer Interaction  
Universität Hamburg

### ABSTRACT

This poster presents an approach to systematically distinguish between interaction techniques (ITs) in the context of magic ITs in immersive virtual environments. Currently, heterogeneous terms are used in research to describe the concept of enhancing the abilities of users beyond the limits of the real world, such as *magic*, *super-natural*, *hyper-natural*, *superhuman abilities*, *superpowers*, *augmentation* or *empowerment*. As a first step towards clarifying and systematically defining the terminology, we propose using the orthogonal concepts of internalizability, congruence, and enhancement (or ICE-cube) as a simple yet expressive conceptual framework.

**Index Terms:** Human-centered computing—Human computer interaction (HCI)—HCI theory, concepts and models; Human-centered computing—Human computer interaction (HCI)—Interaction techniques;

### 1 INTRODUCTION & RELATED WORK

“Any sufficiently advanced technology is indistinguishable from magic” [3] is a popular quote in the context of human-computer interaction. Immersive virtual environments (IVEs), as part of advanced technology, allow the utilization of “magic” interaction techniques (ITs) which are not restricted by the laws of physics or limited by human abilities, such as teleportation, telekinesis or x-ray vision. In the context of magic, compared to other technologies, VR offers enormous possibilities, for example, embodying different avatars, exploring new artificial worlds. Even though magic ITs and similar concepts have been researched for a long time, numerous terms and concepts are used to describe magic ITs and similar ideas in diverse domains, and definitions are vague. This heterogeneity is confusing and inhibits a clear communication about these techniques. In this poster, we propose a unifying descriptive framework as a first step towards a systematic classification naming scheme for ITs.

Various authors, for example, [7, 9, 12, 13], have described the utilization of “magic” ITs in IVEs as an approach to enhance human abilities and overcome limitations of reality or technology by reducing realism. Several terms have been introduced, which describe apparently similar concepts, such as “superhuman” abilities [8], “hyper-” and “super-natural” [10, 14], “augmentation” [16] or “superpowers” [15] — a clear delineation does not exist. Even though some classification systems and definitions have been proposed [4, 10, 14], authors often rely on providing a brief interpretation of used terms instead of referencing a specific conceptual framework.

\*e-mail: bastian.dewitz@studium.uni-hamburg.de

†e-mail: sukran.karaosmanoglu@uni-hamburg.de

‡e-mail: rob.lindeman@canterbury.ac.nz

§e-mail: frank.steinicke@uni-hamburg.de

### 2 A CONCEPTUAL MODEL: ICE-CUBE

We propose a classification system for ITs which can be used to systematically distinguish between ITs. Based on the orthogonal concepts of Internalizability, Congruence, and Enhancement, the **ICE-cube** model is the result of working in the field of magic ITs for several years, and identifying missing definitions and naming conventions as a problem to communicate ideas and relate diverse research projects. Questions emerged, such as: *What are the fundamental properties to consider an IT “magic”?* *Is there a difference between “magic” and “super-natural” and other terms?* The construction of our conceptual model started with the suggestion by one author to combine important properties of magic ITs into a single model to provide a systematic classification system. Following that, we conducted an informal literature review (90 publications) to analyze main components in designing ITs and refine our initial model. The three most mentioned recurrent themes used to describe ITs were (1) *enhancing the abilities* or improving the performance for specific tasks, (2) providing a *high learnability and intuitive use*, and (3) allowing users to perform actions which are *not possible in the real world*, while other relevant components, such as narrative and diegetic elements are less commonly reported. These three main components can be treated as independent dimensions.

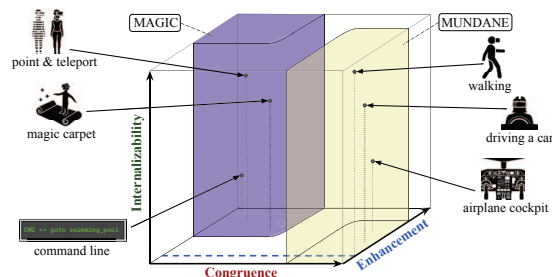


Figure 1: The ICE-cube with the “magic” and “mundane” volumes, and estimated location of some ITs in IVEs.

We use the term **internalizability** to describe the first dimension in our model. Internalization is a term referred to in Activity Theory [6] to describe the process of transforming external operations, such as interpretation of symbols and the use of tools etc., into internal processes. ITs in IVEs can be seen as techno-cultural constructs which need to be integrated into our cognitive structures in the same way to make them intuitively and subconsciously usable. These constructs contain context-dependent factual (“What can I do?”) and associated procedural (“How can I do it?”) knowledge. Internalizability as a property of ITs describes therefore the easiness of internalizing the procedures of an IT into the user’s set of possible means-to-an-end in a specific domain. It is a special form of learnability, in which the techno-culturally invented mapping of simple bodily motor actions to accomplish a (possibly more complex) goal has to be learned. For example, *point & teleport* [2] contains factual knowledge about the application of this IVE-exclusive locomotion technique and specific procedural steps (pointing a controller, pressing & releasing the trigger) which have to be learned. For natural

walking in IVEs, the process of internalization describes the act of learning that real-world walking can be exploited as an IT. We distinguish between internalizability and learnability, as the latter contains other concepts, such as providing tooltips to a novice user, or following guidelines for graphical menus, making it a more general term. One strategy to achieve a high internalizability is the adoption of known metaphors from different domains, or adopting real-world interaction. Without a real-world counterpart, a high internalizability can be achieved by providing an uncomplex and direct mapping between a user input and system action. IVE-exclusive and real-world-adapted ITs can have low internalizability, e.g., using a command-line interface for teleportation, or controlling locomotion with a cockpit of a real-world airplane. Furthermore, internalizability depends on the user; a pilot will internalize the controls of a faithfully recreated airplane in an IVE faster than the average user.

We use the term **congruence** in the sense of the IVE matching the expectations of the real world [1] to refer to intentional modifications to the virtual avatar, altered physics, and also changes in perception and cognition (e.g., providing artificial stimuli). These changes can be recognized on a cognitive level, and compared to real-world experiences. It is closely related to the concept of realism [11] but differs by excluding certain factors (e.g., fidelity-related display resolution). The perceived degree of congruence can differ between individuals with different prior experiences of their perceived reality. Congruence can be treated as a continuum; a magic carpet seems more realistic than teleportation as it involves some incomprehensible device which enables humans to fly, while a car is congruent with our experience of reality, even though most drivers actually do not (need to) fully understand the engineering behind it.

The last dimension in our model is **enhancement**. Enhancing existing skills or enabling new abilities has always been a goal of tool creation and technological development. Similarly, ITs in IVEs as software tools aim at allowing users to perform specific tasks more efficiently, for example, locomotion, object selection or manipulation. It is also possible to enhance the human senses or provide help in cognitive tasks [15]. We can distinguish different degrees of enhancement ranging from exactly transferring the real-world abilities of humans to IVEs (e.g., natural walking) to continuously increasing levels of enhancement (scaled travel speed, “7-league boots” [5]), and, finally, to providing novel ways of interaction (flying, teleportation). Relating enhancement to the typical potential of a healthy adolescent to perform a task without technological support (thick dashed line in Fig. 1) provides a reasonable baseline.

ITs can be assessed with these dimensions to obtain a location within the ICE-cube in which ITs with similar properties are in close proximity to each other. Measuring these dimensions and determining the threshold values for ITs which need to be exceeded to classify an IT as “magic” is, however, challenging, as we assume high subjective differences. However, from a conceptual standpoint, crossing these borders seems necessary for magic ITs as the coherent definition of magic ITs is the alteration of reality (low congruence) to exceed human capabilities or overcome limits of reality and technology (high enhancement) [12, 13]. Crossing these borders is visible as a distinct “magic” volume within the ICE-cube (see Fig. 1) containing ITs with both high and low internalizability. On the opposite side, the “mundane” [13] volume can be located. It describes ITs which faithfully recreate the real-world potential for performing a task (e.g., real-world walking or driving a car as a real-world congruent enhancement). Distinct sections of the volume of the ICE-cube can be interpreted as classes of ITs. For example, *super-natural ITs* can be located at high internalizability, low congruence and high enhancement, following previously published definitions [4, 10, 14], which makes them a sub-class of magic ITs. We believe that other research terms can be described using distinct volumes within the ICE-cube. This is, however, up for discussion and part of the future work concerning the classification of ITs.

### 3 CONCLUSION & FUTURE WORK

In this work, we introduced the ICE-cube. We see this model as a promising approach to creating a systematic naming scheme and classification for ITs. We believe that providing a consistent and precise naming scheme facilitates communicating ideas and concepts between heterogeneous groups and within interdisciplinary teams.

Magic ITs may become more important in our everyday interaction in the near future considering the current technological steps in the fields of virtual and augmented reality. Novel means of interaction with powerful computer systems offer the potential to transform the way we work and live, just as smartphones began to revolutionize our society 15 years ago, and personal computers did so 50 years ago. Interaction with these devices which would have been perceived as “magic” some decades earlier is now part of our everyday lives. With a look at Clark’s *Profiles of the future* [3], one can put it this way: *Any sufficiently established advanced technology becomes mundane.*

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